Based on the content of both CMP510 question papers you uploaded, here are **the top 20 most important questions** selected for their conceptual depth, practical relevance, and recurrence in exams:

Top 20 Most Important Questions (CMP510 - Computer Hardware)

1. **Explain in detail North/South bridge architecture of chipset.**
2. **Describe Modified Frequency Modulation (MFM) and Run Length Limited (RLL) techniques of recording with suitable example.**
3. **Write differences between CRT, LCD and LED.**
4. **What are the types of Power Supply? Explain in detail.**
5. **State differences between IDE, SATA and SCSI.**
6. **List out the Components of motherboard. Explain in detail any one of them.**
7. **Describe Virtual Real (8086) mode of processor.**
8. **Explain working principle of Hard Disk with neat diagram.**
9. **Construction and working of Touch Screen Technologies.**
10. **Draw the block diagram of RS232 connector and give the function of the signals.**
11. **Write short notes on UPS and Bluetooth.**
12. **Write a short note on CPU registers.**
13. **What is cache memory? Define cache hit and cache miss.**
14. **Explain in detail NTFS file system.**
15. **Explain working of SMPS with neat diagram.**
16. **Define UPS. Explain its types.**
17. **What are the symptoms of power supply problem?**
18. **Explain the differences between FAT16 and FAT32 file systems.**
19. **Define POST. Explain POST sequence.**
20. **Explain applications of Parallel Processing.**

Let me know if you'd like model answers for any or all of these!

1. Explain in detail North/South bridge architecture of chipset.

- **The chipset on a motherboard is divided into two parts: North Bridge and South Bridge, historically used in older PCs.**
- **North Bridge:**
  - **Connects high-speed components like CPU, RAM, and graphics card.**
  - **Handles communications between processor and memory (RAM).**
  - **Often includes memory controller and graphics controller.**
- **South Bridge:**
  - **Connects slower I/O devices such as USB, audio, storage (HDD/SSD), and network interfaces.**
  - **Responsible for managing BIOS, CMOS, and peripheral connectivity.**
- **Working:**
  - **CPU communicates with RAM and GPU via North Bridge.**

    - **South Bridge interacts with North Bridge for data that involves I/O or peripheral devices.**
- **In modern systems, Intel's Platform Controller Hub (PCH) combines both bridges into a single chip to improve performance and reduce latency.**

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2. Describe MFM and RLL techniques of recording with suitable example.

- **Modified Frequency Modulation (MFM):**
    - **Used in older hard drives and floppy disks.**
    - **A data bit is recorded using a specific timing and frequency of magnetic changes.**
    - **Stores less data per inch compared to newer techniques.**
- **Run Length Limited (RLL):**
    - **An advanced method that encodes sequences with fewer transitions.**
    - **Increases storage capacity without changing physical medium.**
    - **More efficient than MFM, allows 50% more data per track.**
- **Example:**
    - **If MFM stores 17 sectors/track, RLL can store 26 sectors/track.**
- **RLL requires advanced controller circuitry but provides higher performance and storage density.**

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3. Write differences between CRT, LCD and LED.

Feature	CRT	LCD	LED
Technology	Electron gun with phosphor	Liquid crystals	LCD with LED backlighting
Size & Weight	Bulky and heavy	Thin and light	Slimmer and more efficient
Image Quality	Sharp, but flickers	Clearer, less strain	Brighter, better contrast
Power Consumption	High	Medium	Low
Lifespan	Shorter	Longer	Longest
Applications	Old TVs, monitors	Desktop monitors	TVs, laptops, smartphones

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4. What are the types of Power Supply? Explain in detail.

- **1. Linear Power Supply:**
    - **Simple design, uses transformers.**
    - **Produces less noise but is bulky and inefficient.**
- **2. Switched Mode Power Supply (SMPS):**
    - **Converts AC to DC using high-frequency switching.**

    - **Compact, efficient, and widely used in computers.**
- **3. Uninterruptible Power Supply (UPS):**
    - **Backup power in case of main power failure.**
    - **Includes batteries and inverters.**
    - **Types: Online, Offline, Line-interactive.**
- **4. Modular Power Supply:**
    - **User connects only needed cables.**
    - **Improves airflow and aesthetics.**
- **5. External Power Adapters:**
    - **Used in laptops and small electronics (AC to DC conversion).**

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5. State differences between IDE, SATA and SCSI.

Feature	IDE (PATA)	SATA	SCSI
Speed	Up to 133 MB/s	150–600 MB/s	Up to 640 MB/s or more
Cable Type	40/80-pin ribbon	7-pin thin cable	50/68/80-pin ribbon
Max Devices/Bus	2	1	Up to 16
Cost	Cheap	Moderate	Expensive
Hot Swappable	No	Yes	Yes
Use Cases	Old PCs	Modern PCs, laptops	Servers, enterprise systems

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6. List out components of motherboard. Explain any one in detail.

- **Components:**
    - **CPU Socket**
    - **RAM Slots**
    - **BIOS/UEFI Chip**
    - **Chipset (North & South Bridge)**
    - **PCI/PCIe Slots**
    - **SATA Connectors**
    - **Power Connectors**
    - **CMOS Battery**
    - **USB/Audio Ports**
- **BIOS/UEFI Chip (detailed):**
    - **Stores firmware required to start the system.**
    - **Performs POST (Power-On Self-Test).**
    - **Allows configuration of hardware settings.**
    - **UEFI is the modern version, supporting large drives and graphical UI.**

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7. Describe Virtual Real (8086) mode of processor.

- **Introduced in 80386 and higher Intel processors.**
- **Allows execution of real-mode (8086) applications in protected mode.**
- **Uses paging to provide memory protection and multitasking.**
- **Each real-mode program runs in its own virtual 8086 environment.**
- **Useful for running DOS applications in a multitasking OS like Windows.**

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8. Explain working principle of Hard Disk with neat diagram.

- **Platters: Magnetic disks where data is stored.**
- **Spindle Motor: Spins the platters at high speeds (e.g., 7200 RPM).**
- **Read/Write Heads: Positioned on actuator arms, read or write magnetic signals.**
- **Actuator Arm: Moves the head precisely over the disk.**
- **Working:**
  - **Data is stored as magnetic patterns on platters.**
  - **Heads float on air cushion close to platters.**
  - **Sectors and tracks organize data for access.**
- **Diagram includes: Platters, spindle, actuator arm, read/write head.**

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9. Construction and working of Touch Screen Technologies.

- **1. Resistive:**
  - **Two layers with electrical resistance.**
  - **Touch creates contact, changing voltage.**
  - **Works with stylus or gloved finger.**
- **2. Capacitive:**
  - **Uses human conductivity to detect touch.**
  - **Allows multi-touch, high durability.**
- **3. Infrared:**
  - **Uses IR beams across the screen.**
  - **Touch interrupts the beams, determining position.**
- **4. SAW (Surface Acoustic Wave):**
  - **Uses ultrasonic waves.**
  - **Touch absorbs part of wave to detect location.**

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10. Block diagram of RS232 and signal functions.

- **RS232: Standard for serial data transmission.**
- **Block Diagram Elements:**
  - **DB9 Connector**
  - **TXD (Transmit Data)**
  - **RXD (Receive Data)**

    - **RTS (Request to Send)**
    - **CTS (Clear to Send)**
    - **DSR (Data Set Ready)**
    - **DTR (Data Terminal Ready)**
    - **GND (Ground)**
- **Functions:**
    - **Ensures reliable serial communication between PC and external devices like modems.**

11. Write short notes on UPS and Bluetooth

a) UPS (Uninterruptible Power Supply):

- **A UPS is a backup device that supplies power to a computer when the main power fails.**
- **It prevents data loss and damage to equipment.**
- **Types:**
    - **Offline UPS: Starts working only when power fails.**
    - **Online UPS: Always provides power through the battery; no delay.**
    - **Line-Interactive UPS: Maintains voltage and provides backup during failure.**
- **Common in hospitals, banks, and data centers.**

b) Bluetooth:

- **Bluetooth is a wireless technology for short-range communication (up to 10 meters).**
- **Works on 2.4 GHz frequency.**
- **Used to connect devices like mobile phones, headphones, speakers, and keyboards.**
- **Saves energy and allows easy pairing without cables.**

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12. Write a short note on CPU registers

- **Registers are small, fast storage units inside the CPU.**
- **Used to store data, instructions, and memory addresses during execution.**
- **Types of Registers:**
    - **General Purpose: Used for arithmetic and logic (e.g., AX, BX).**
    - **Control Registers: Manage program execution (e.g., Instruction Pointer).**
    - **Status Registers: Store information about results of operations (e.g., Zero or Carry flags).**
- **Registers work faster than RAM and help the CPU perform tasks efficiently.**

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13. What is cache memory? Define cache hit and cache miss

- **Cache Memory: A very fast memory located near the CPU that stores frequently used data.**
- **Cache Hit: When the CPU finds the required data in the cache → fast access.**
- **Cache Miss: When the data is not found in cache → it is fetched from slower RAM.**
- **Cache improves the overall speed of the computer system.**

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14. Explain in detail NTFS file system

- **NTFS (New Technology File System): Used by Windows operating systems.**
- **Features:**
  - **Supports large files and large hard drives.**
  - **Provides security with file permissions and encryption.**
  - **Supports file compression to save space.**
  - **Keeps a log of changes to help recover data after a crash.**
- **NTFS is more advanced and secure than older file systems like FAT.**

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15. Explain working of SMPS with neat diagram

- **SMPS (Switched Mode Power Supply) converts high-voltage AC to low-voltage DC.**
- **Working Steps:**
  1. **AC input is converted to DC using a rectifier.**
  2. **DC is switched on and off rapidly to create high-frequency signals.**
  3. **These signals go through a transformer to adjust voltage.**
  4. **Output is converted back to smooth DC and given to devices.**
- **SMPS is efficient, compact, and used in computers and TVs.**

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16. Define UPS. Explain its types

- **Definition: A UPS is a power backup device that protects devices from power failure.**
- **Types:**
  - **Offline UPS: Starts supplying power only when main power fails.**
  - **Online UPS: Always supplies power through battery, no delay.**
  - **Line-Interactive UPS: Corrects minor voltage problems without switching to battery.**
- **Helps in preventing data loss and equipment damage during power issues.**

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17. What are the symptoms of power supply problem?

- **Computer does not start at all.**
- **Sudden restart or shutdown during use.**
- **Burning smell or strange noise from the power supply.**
- **Computer crashes randomly or behaves abnormally.**
- **Fans or lights may not work.**
- **All these show the power supply might be faulty and needs checking.**

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18. Explain the differences between FAT16 and FAT32

Feature	FAT16	FAT32
Max File Size	2 GB	4 GB
Max Partition Size	2 GB	2 TB
Cluster Size	Larger (wastes space)	Smaller (efficient)
Speed	Slower	Faster
Compatibility	Works on older systems	Better for modern systems

- **FAT32 is an improved version of FAT16 with better space use and support for larger drives.**

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19. Define POST. Explain POST sequence

- **POST (Power-On Self-Test) is a process run by the computer when it is turned on.**
- **It checks if the hardware is working properly (RAM, keyboard, hard disk, etc.).**
- **Sequence:**
    1. **Computer is powered on.**
    2. **CPU runs the BIOS program.**
    3. **BIOS checks hardware (RAM, display, etc.).**
    4. **If hardware is fine, a single beep sound is heard.**
    5. **If there's a problem, error codes or beeps are given.**
    6. **After POST, the operating system starts loading.**
- **POST helps detect and report hardware issues at startup.**

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20. Explain applications of parallel processing

- **Parallel Processing: Using multiple processors or cores to perform tasks at the same time.**
- **Applications:**
    - **Scientific Research: Simulating weather, earthquakes, or space.**
    - **Medical Imaging: Faster scanning and diagnosis.**
    - **Gaming: Smooth graphics and fast processing.**
    - **Artificial Intelligence (AI): Faster training of models.**
    - **Data Analysis: Processing large amounts of data in less time.**
- **It improves speed and performance in tasks that require a lot of computing power.**